杨奕铭 Yiming Yang

yimingyang@link.cuhk.edu.cn (+86) 152 2064 6049
香港中文大学（深圳）
谷歌学术 个人主页

教育与学术访问

南洋理工大学，新加坡 2026.10 – 2027.10（进行中）
访问博士生，AutoMan Lab
合作导师：吕辰教授
国家留学基金管理委员会（CSC）资助，2026 年 5 月获批

香港中文大学（深圳） 2024.08 – 至今
计算机与信息工程博士候选人
导师：李镇教授、崔曙光教授
全额奖学金；2026 年 7 月通过博士资格考试（QE）

中山大学 2021.08 – 2024.06
电子与通信工程硕士
综合测评排名第 1
导师：胡海峰教授

西北农林科技大学 2017.08 – 2021.06
电子信息工程学士
GPA：3.85/4.00（前 2.94%）

研究方向

自动驾驶、车道拓扑推理、世界模型、跨模态行人重识别与人脸识别

研究论文与预印本

注：论文题目与作者保留英文，便于检索和引用。

[1] Yueru Luo, Changqing Zhou, **Yiming Yang**, Erlong Li, Chao Zheng, Shuguang Cui, Zhen Li.
Geometry-Guided Representations for Coherent Lane and Traffic Topology Reasoning in Driving Scenes
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2026.

[2] Hongbin Lin, Chaoda Zheng, **Yiming Yang**, Xiangyu Li, Shijia Chen, Jinhao Deng, Kangjie Chen, Dongbin Zhang, Jie Feng, Yu Zhang, Xianming Liu, Shuguang Cui, Boyang Wang, Zhen Li.
RoXDrive: Closed-Loop Reinforcement Learning for End-to-End Autonomous Driving via Action-Faithful Rollouts
arXiv 预印本, 2026.

[3] **Yiming Yang**, Yueru Luo, Bingkun He, Erlong Li, Zhipeng Cao, Chao Zheng, Shuqi Mei, Zhen Li.
Topo2Seq: Enhanced Topology Reasoning via Topology Sequence Learning
AAAI Conference on Artificial Intelligence (**AAAI Oral**), 2025.

[4] **Yiming Yang**, Weipeng Hu, Haifeng Hu.
Dynamic Modality-Camera Invariant Clustering for Unsupervised Visible-Infrared Person Re-Identification
IEEE Transactions on Neural Networks and Learning Systems (**T-NNLS**), 2025.

[5] **Yiming Yang**, Weipeng Hu, Haifeng Hu.
Progressive Cross-Modal Association Learning for Unsupervised Visible-Infrared Person Re-Identification
IEEE Transactions on Information Forensics and Security (**T-IFS**), 2025.

[6] **Yiming Yang**, Weipeng Hu, Haifeng Hu.
Unsupervised NIR-VIS Face Recognition via Homogeneous-to-Heterogeneous Learning and Residual-invariant Enhancement
IEEE Transactions on Information Forensics and Security (**T-IFS**), 2023.

[7] **Yiming Yang**, Weipeng Hu, Haiqi Lin, Haifeng Hu.
Robust Cross-domain Pseudo-labeling and Contrastive Learning for Unsupervised Domain Adaptation NIR-VIS Face Recognition
IEEE Transactions on Image Processing (**T-IP**), 2023.

[8] **Yiming Yang**, Weipeng Hu, Haifeng Hu.
Syncretic Space Learning Network for NIR-VIS Face Recognition
ACM Transactions on Multimedia Computing, Communications, and Applications (**T-OMM**), 2023.

[9] **Yiming Yang**, Weipeng Hu, Haifeng Hu.
Neutral Face Learning and Progressive Fusion Synthesis Network for NIR-VIS Face Recognition
IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2023.

[10] Hongbin Lin, **Yiming Yang**, Chaoda Zheng, Yifan Zhang, Shuangcheng Niu, Zilu Guo, Yafeng Li, Gui Gui, Shuguang Cui, Zhen Li.
DriveFlow: Rectified Flow Adaptation for Robust 3D Object Detection in Autonomous Driving
AAAI Conference on Artificial Intelligence (**AAAI**), 2026.

[11] Qiaolin He, **Yiming Yang**, Zihan Wang, Haifeng Hu.
Part-based Bi-directional Enhancement Learning for Unsupervised Visible-Infrared Re-identification
IEEE Transactions on Information Forensics and Security (**T-IFS**), 2025.

[12] Jiakai He, **Yiming Yang**, Ruixing Wu, Haifeng Hu.
Unsupervised Visible-Infrared Person ReID via Modality-Camera Balance Label Refinement
ACM Transactions on Multimedia Computing, Communications, and Applications (**T-OMM**), 2025.

[13] Ruixing Wu, **Yiming Yang**, Jiakai He, Haifeng Hu.
Extended Cross-Modality United Learning for Unsupervised Visible-Infrared Person Re-identification
IEEE Signal Processing Letters (**SPL**), 2025.

[14] Suzhong Fu, Rui Sun, Xuan Ding, Jingqi Dong, **Yiming Yang**, Yao Zhu, Min Chang Jordan Ren, Delin Deng, Angelica Aviles-Rivero, Shuguang Cui, Zhen Li.
VesSAM: Efficient Multi-Prompting for Segmenting Complex Vessel
International Conference on Bioinformatics and Biomedicine (**BIBM**), 2025.

[15] Weipeng Hu, Jiun Tian Hoe, Runzhong Zhang, **Yiming Yang**, Haifeng Hu, Yap-Peng Tan.
Motion-guided token prioritization and semantic degradation fusion for exo-to-ego cross-view video generation
Information Fusion, 2025.

[16] Weipeng Hu, **Yiming Yang**, Haifeng Hu.
Pseudo Label Association and Prototype-based Invariant Learning for Semi-Supervised NIR-VIS Face Recognition
IEEE Transactions on Image Processing (**T-IP**), 2024.

在审论文

[17] Yihuan Yao*, **Yiming Yang***, Hongbin Lin, Chuang Yang, Chao Zhan, Suzhong Fu, Chao Zheng, Chen Lv, Shuguang Cui, Zhen Li.
TopoMind: Unifying Lane Topology Reasoning and Understanding on Driving Scenes
在审. * 共同第一作者.

[18] **Yiming Yang**, Chao Zhan, Yihuan Yao, Hongbin Lin, Yueru Luo, Wan lihui, Suzhong Fu, Chao Zheng, Chen Lv, Shuguang Cui, Zhen Li.
TopoVLM: Unveiling the Power of Vision-Language Models in Lane Topology Reasoning
在审.

[19] **Yiming Yang**, Hongbin Lin, Yueru Luo, Suzhong Fu, Chao Zheng, Xinrui Yan, Shuqi Mei, Kun Tang, Chen Lyu, Shuguang Cui, Zhen Li.
DualTopo: Spatio-Temporal Lane Segment Graph Generation with Dual Latent World Models and Parallel Query Supervision
在审.

[20] **Yiming Yang**, Yueru Luo, Bingkun He, Hongbin Lin, Suzhong Fu, Chao Zheng, Zhipeng Cao, Erlong Li, Chao Yan, Chen Lyu, Shuguang Cui, Zhen Li.
TopoX: Spatio-Temporal Lane Graph Generation via Iterative Positional Embedding and Denoising
在审.

[21] Suzhong Fu, Yizhou Yang, Jing Wang, Jingqi Dong, Xuan Ding, **Yiming Yang**, Shuguang Cui, Zhen Li.
CLSim: Lesion-Centric Coronary Stenosis Editing with Geometry Coupled Supervision
在审.

[22] Chao Zhan, **Yiming Yang**, Chao Yan, Bin Wang, Chao Zheng, Shuguang Cui, Zhen Li.
MapDLM: Geometry-Grounded Block-Diffusion Vision-Language Models for Lane Topology Generation
在审.

[23] Chao Zhan, **Yiming Yang**, Chao Yan, Chao Zheng, Shuguang Cui, Zhen Li.
LaneSpatial: Benchmarking VLM Spatial Understanding of BEV Road-Map Structure
在审.

荣誉与奖励

- 国家公派留学奖学金（CSC），2026 年 5 月
- 国家奖学金（2018、2020、2023）
- 研究生奖学金（2021、2022、2023）
- 一等奖学金（2018、2019、2020）
- 中山大学优秀硕士毕业生、优秀硕士学位论文（2024）
- 西北农林科技大学优秀本科毕业论文（2021）

教学经历

- 助教，CIE6020：信息论，香港中文大学（深圳），2025 年秋季
- 助教，CIE6032：深度学习基础与应用专题，香港中文大学（深圳），2026 年夏季

学术服务

研讨会组织

- 组织者，*TopoCoT*：通过思维链设计实现稳健、可泛化的车道拓扑理解与高精地图生成，
  IEEE 计算机视觉应用冬季会议研讨会 (**WACV**), 2026
  提案作者: **Yiming Yang**, Chao Zheng, Hongyang Li, Zhen Li.
  官网

期刊审稿

- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Information Forensics and Security
- IEEE Transactions on Circuits and Systems for Video Technology
- Pattern Recognition
- Engineering Applications of Artificial Intelligence
- Digital Signal Processing

会议审稿

- AAAI Conference on Artificial Intelligence (AAAI 2025–2027)
- Conference on Neural Information Processing Systems (NeurIPS 2025, 2026)
- International Conference on Computer Vision (ICCV 2025)
- European Conference on Computer Vision (ECCV 2026)
- International Conference on Learning Representations (ICLR 2026, 2027)
- Conference on Computer Vision and Pattern Recognition (CVPR 2026)
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026)

专业技能

- **编程与工具**: Python, MATLAB, PyTorch, TensorFlow, OpenCV, Linux, LaTeX
- **语言能力**: 英语（雅思 7.0、大学英语六级）